LOT PAGER ??? 01-PLAN-AND-ROADMAP

LOT® PAGER (HW-01) — Plan & Roadmap

Owner: Vadik Marmeladov, Inventor · COSMO® CIA · Version 0.1 · 2026-10-17 Status: PLAN — nothing ordered, nothing fabricated.

1. What it is

A 40 × 40 mm flat stainless-steel object that receives short autonomous notifications from lot-systems.com (“Coffee time!”) on a small screen, and sends a signal back to the site’s **Log** tab when its single button (“Copy”) is pressed. Camera and weather sensing make it an input device for the LOT® system, not only an output.

- Side A: polished stainless steel, blank.
- Side B: screen, camera, button.
- Charging: wireless (Qi). No port. Sealed.

Lineage: CUBIQ™ (docs/corporate/LOT-CUBIQ-QUANTUM-CUBE-v0.md) is the haptic notification body; COSMO® Cube (LOT-MANIFEST, brave-lamport-t9z5u8) is the general-purpose computer. LOT® PAGER sits between them: screen + camera + sensors, no actuated motion. It does not reuse the name CUBIQ™ or Cube.

2. Requirement trace (your 19 points)

#	Requirement	Where addressed
1	PCBWay	04-ELECTRONICS-PCBWAY, 07-PRODUCTION-100
2	Pager-like notification from AI site	06-API-CONNECTOR §3
3	2-part stainless body	03-MECHANICAL
4	Flat silver square 4×4 cm × 5 mm	03-MECHANICAL §2 (**5 mm not feasible in v0 — see §4**)
5	Camera	02-BOM, 05-FIRMWARE §6
6	LOT API connector	06-API-CONNECTOR
7	PDF manuals	build.sh → /pdf (generated)
8	Compress info per session	08-SESSION-PROTOCOL
9	Firmware documents	05-FIRMWARE
10	Software to connect with firmware	tools/pager-sim.mjs, 06
11	Separate documents	one file per discipline (this folder)

#	Requirement	Where addressed
12	Charger	02-BOM §Charging, 03 §5
13	100-unit run	07-PRODUCTION-100
14	Weather sensor	BME688 (T/RH/P/gas)
15	AI-grade off-the-shelf sensors	02-BOM §Sensors
16	Copy button → Log tab	06 §4 (uses existing POST /api/logs)
17	Polished stainless side	03 §3
18	Camera + screen + button side	03 §2
19	Wireless charger	02-BOM, 03 §5

3. Phased roadmap

Phase	Weeks	Output	Gate to next phase
P0 Dev kit	0–3	Off-the-shelf ESP32-S3 + 1.3″ IPS + BME688 on a breadboard; pager-sim.mjs talks to a staging LOT server; Copy writes a Log entry	Notification → screen → button → Log works end-to-end
P1 Server	2–5	Device endpoints (06 §3) behind a feature flag; device tokens; AI notification generator	Staging green
P2 Board v0.1	4–10	4-layer PCB, 2 spin of 5 boards at PCBWay; sizes to 40×40 mm outline	Board boots, Wi-Fi, screen, BME688, Qi charge ok
P3 Enclosure EVT	8–14	10 CNC 316L two-part shells at ~8–9 mm; polished	Fit check, RF check through steel, thermal check
P4 Camera/AI	10–16	Camera capture → upload → server-side AI; privacy shutter policy	Capture works inside enclosure
P5 DVT	14–22	25 units, full firmware, OTA, drop/ingress tests, FCC/CE pre-scan	Pre-scan pass
P6 Pilot 100	22–34	100-unit build (07)	Yield ≥ 90 %, burn-in pass
P7 Slim v1	30+	Chase the 5 mm target with flex PCB + custom camera	Only if P6 sells the idea

Total to 100 units: **~8 months** from a funded start. Dominant risk is enclosure + RF, not electronics.

4. Key finding: 4×4 cm × 5 mm is not achievable in v0

Stack inside a 5 mm body with 2 × ~0.5 mm steel walls leaves ~4 mm:

Layer	Thickness
Screen module (IPS 1.3″, incl. glass+FPC)	~1.5–2.0
PCB + tallest parts (ESP32-S3-MINI-1 is 2.4)	~3.2
Li-Po (thinnest practical, e.g. 301830-class)	~3.0
Qi coil (flex, w/ ferrite)	~0.5–0.8
Camera (smallest common DVP modules are 5–6 mm tall)	~5+

Even before steel the sum is >10 mm if stacked. Realistic targets:

- **v0/P3: 40 × 40 × 9 mm** (what we can actually build).
- **v1: ~40 × 40 × 6.5 mm** with side-by-side layout, 1.54″ e-paper or thinner screen, wafer camera, flex PCB — needs custom parts and a larger NRE.
- **5 mm** requires dropping the camera or the battery (tethered/charged-pad operation). Decision needed from Vadik — see 09-RISKS D1.

5. Decisions needed from Vadik

1. Accept 9 mm for v0, or drop camera/battery to chase 5 mm?
2. Screen: IPS colour (richer, ~60–80 mA) vs. e-paper (pager feel, days of battery, slow refresh)?
3. “Copy” copies what? (Assumed: marks the current notification as acknowledged and logs it; nothing goes to the clipboard of a host.)
4. Camera privacy: physical shutter vs. LED + log-everything policy.
5. Budget ceiling for the 100-unit run (07 estimates $12–20k).